# FUNCTIONAL REQUIREMENT DOCUMENT (FRD)
# XR Simulation Learning Platform

## SECTION 1 – PLATFORM OVERVIEW

This section defines the fundamental characteristics of the platform including its architecture model, supported simulation formats, content governance structure, and multi-tenant system design.
The goal of this section is to establish the **core structural rules of the platform before functional features are described.**

### Phase 1 – Core Platform Architecture

Phase 1 establishes the foundational platform architecture including the multi-tenant structure, supported simulation environments, and the content governance workflow.

### 1.1 Platform Model

#### 1.1.1 Multi-Tenant SaaS Platform

The platform shall operate as a **multi-tenant SaaS-based Learning Management System (LMS)** designed to distribute XR-based engineering simulation modules to both institutions and individual learners.

#### 1.1.2 B2B Institutional Learning Environment

The system shall support institutional deployments where organizations purchase licenses and allow students and faculty within the institution to access the platform under a shared institutional subscription.

#### 1.1.3 B2C Individual Learning Environment

The platform shall support individual learners who independently purchase modules or learning packages and access simulations directly through personal accounts.

#### 1.1.4 Centralized Content Ownership

All XR modules hosted on the platform shall remain under the ownership and governance of the platform owner and shall not be modifiable or redistributable by institutions.

### 1.2 Supported Simulation Formats

#### 1.2.1 Web Simulation Support

The platform shall support browser-based simulations that can run directly within standard web environments without requiring XR hardware.

#### 1.2.2 VR Simulation Support

The platform shall support virtual reality simulations deployed through VR headsets such as Meta Quest devices.

#### 1.2.3 MR Simulation Support

The platform shall support mixed reality simulations designed for MR hardware such as Ajna Lens.

#### 1.2.4 AR Simulation Support

The platform shall support augmented reality simulations that can run on mobile devices using Android and iOS operating systems.

### 1.3 Multi-Tenant System Isolation

#### 1.3.1 Tenant-Wise Database Isolation

Each institution's data shall be stored in logically separated tenant environments to prevent cross-access between institutions. This isolation shall apply to user records, analytics data, licensing data, and institution-specific course mapping.

#### 1.3.2 B2C Database Isolation

B2C users shall be maintained in a separate database environment that is independent of institutional databases to ensure clean separation between consumer and institutional data.

**1.3.3 B2C Database Expansion**

The platform shall support deployment of additional B2C database instances across different geographic regions to ensure system performance and reduce latency for large user clusters.

**1.3.4 Tenant-Scoped Analytics**

Analytics and reporting generated for each institution shall be limited to that tenant's dataset and shall not include data from other institutions.

**1.3.5 Cross-Tenant Governance Access**

A Super Admin role shall have controlled access to view and manage data across tenants strictly for governance and operational oversight.

**1.4 Content Validation & Publishing Workflow**

**1.4.1 Vendor Module Submission**

Authorized vendors shall submit XR module builds and associated assets through the platform's staging environment for review.

**1.4.2 Academic Validation by BITS Pilani**

All submitted modules shall undergo academic validation by the BITS Pilani academic team to ensure alignment with engineering curriculum standards.

**1.4.3 Content Manager Approval**

Following academic validation, the Content Manager shall review metadata, documentation, and platform compatibility before approving the module for publication.

**1.4.4 Production Publishing by Super Admin**

Only the Super Admin shall have permission to publish approved modules to the production environment.

**1.4.5 Restricted Vendor Publishing**

Third-party vendors shall not have direct publishing rights to the production platform.

**Phase 2 – Content Lifecycle Governance**

Phase 2 introduces structured content lifecycle management and version control to support long-term academic stability of simulation modules.

**1.5 Content Lifecycle Management**

**1.5.1 Module Lifecycle States**

Each XR module shall maintain lifecycle states including Draft, Under Review, Approved, Published, and Archived.

**1.5.2 Academic Version Locking**

Institutions shall be able to lock a specific version of a module during an academic semester to ensure learning continuity.

**1.6 Module Version Management**

**1.6.1 Version Tracking**

The platform shall maintain version history for each XR module to track updates and revisions.

**1.6.2 Version Rollback**

Administrators shall be able to revert modules to earlier versions in case of technical or academic issues.

**Phase 3 – Global Platform Scalability**

Phase 3 focuses on infrastructure capabilities required for global platform expansion.

### 1.7 Regional Platform Deployment

#### 1.7.1 Regional CDN Distribution

XR asset bundles shall be distributed through regional CDN nodes to reduce download latency.

#### 1.7.2 Multi-Region Database Replication

The platform shall support multi-region database replication to ensure availability and performance across global deployments.

## SECTION 2 – XR CONTENT DELIVERY ARCHITECTURE

This section defines how XR simulation modules are packaged, deployed, updated, and controlled across supported XR devices. The architecture ensures secure content distribution, centralized update management, and device-level license enforcement.

### Phase 1 – Core XR Delivery Infrastructure

Phase 1 establishes the architecture required to securely deliver XR simulations to supported devices.

#### 2.1 XR Application Architecture

##### 2.1.1 Master VR Application

A centralized VR application shall be deployed through the Meta Quest Store which will act as the container for downloading and launching XR simulation modules.

##### 2.1.2 MR Device Application

Mixed Reality simulations shall run through a sideloaded application deployed on Ajna Lens devices.

#### 2.2 Secure Simulation Packaging

##### 2.2.1 XR Asset Bundle Packaging

Simulation modules shall be packaged as encrypted asset bundles that are dynamically downloaded by the master XR application.

##### 2.2.2 License-Based Module Access

Simulation modules shall only be downloaded after license validation is confirmed by the platform.

#### 2.3 Module Update Management

##### 2.3.1 Automatic Module Updates

The XR application shall automatically download updated versions of simulation modules when available.

##### 2.3.2 Controlled Update Releases

Super Admin shall control when new versions of modules are released to production environments.

##### 2.3.3 Module Update Without Application Reinstallation

Simulation modules shall be updated independently of the master XR application to avoid requiring full application reinstallation.

#### 2.4 XR Device License Enforcement

##### 2.4.1 B2C Device Binding

Individual user licenses shall be bound to a specific encrypted XR device ID and a user account.

##### 2.4.2 Device Replacement Authorization

Users requesting device replacement must receive approval from the Super Admin before transferring their license to a new device.

##### 2.4.3 Institutional Device Access

Institutional deployments shall allow multiple students to access simulations on shared XR devices.

##### 2.4.4 Session-Based Device Login

Students in institutional environments must log into the XR application using their LMS credentials to access licensed modules.

### Phase 2 – XR Device Management

Phase 2 introduces tools for managing XR devices deployed in institutional labs.

#### 2.5 XR Device Registration

##### 2.5.1 Device Registration

Institutions shall be able to register XR devices on the platform to track device ownership and usage.

**2.5.2 Device Metadata Tracking**

The system shall record device metadata including device ID, device type, institution, and lab location.

**2.6 Remote Device Configuration**

**2.6.1 Remote Module Control**

Administrators shall be able to enable or disable modules on specific XR devices remotely.

**2.6.2 Remote Application Updates**

XR application updates may be pushed remotely to registered devices.

**Phase 3 – XR Telemetry & Interaction Analytics**

Phase 3 introduces deeper tracking of XR interactions for learning analytics.

**2.7 XR Telemetry Tracking**

**2.7.1 Simulation Interaction Logs**

The platform shall capture interaction logs from XR sessions to analyze how learners interact with simulations.

**2.7.2 Simulation Error Tracking**

The system shall record errors encountered by users during simulation scenarios to identify common learning difficulties.

## SECTION 3 – CONTENT ORGANIZATION & LEARNING STRUCTURE

This section defines how learning content is structured on the platform and how modules, courses, and certification programs are organized.

### Phase 1 – Core Learning Structure

Phase 1 introduces the hierarchical content model used by the platform.

#### 3.1 Module Architecture

**3.1.1 Simulation Module Definition**

A module shall represent the smallest learning unit and shall contain a single XR simulation experience.

**3.1.2 Module Learning Components**

Modules may include supporting learning material such as theoretical explanations, simulation instructions, and assessments.

#### 3.2 Course Structure

**3.2.1 Course Composition**

A course shall consist of a collection of modules grouped around a specific engineering topic or learning objective.

**3.2.2 Course Module Sequencing**

Courses shall define the recommended sequence in which modules must be completed.

#### 3.3 Certification Programs

**3.3.1 Certification Program Structure**

Certification programs shall consist of one or more courses grouped into a structured learning pathway.

**3.3.2 Certification Completion Criteria**

Students shall receive certification upon completing all required courses and meeting minimum assessment performance thresholds.

### Phase 2 – Institutional Course Configuration

Phase 2 introduces institutional customization of course structures.

#### 3.4 B2B Course Configuration

**3.4.1 Institutional Course Creation**

Institute Admins shall be able to create courses by grouping available simulation modules.

**3.4.2 Course Module Mapping**

Institute Admins shall be able to add or remove modules from courses within their tenant environment.

**3.4.3 Course Stream Assignment**

Courses shall be mapped to specific academic streams or departments within an institution.

### Phase 3 – Certification Tracking

Phase 3 introduces certification progress.

#### 3.5 Certification Tracking

**3.5.1 Student Certification Progress**

Students shall be able to track progress toward certification programs through their dashboard.

**3.5.2 Certification Eligibility Reports**

Faculty and institutional administrators shall be able to view reports identifying students who have met certification eligibility criteria.

**SECTION 4 – USER MANAGEMENT & ONBOARDING**

This section defines how users join the platform, configure their profiles, and are assigned learning entitlements.

**Phase 1 – Core User Access**

**4.1 Individual User Registration**

**4.1.1 Email Registration**

Users shall be able to create accounts using email registration.

**4.1.2 Social Login**

Users may register using supported social login providers.

**4.2 Institutional User Onboarding**

**4.2.1 Institutional Account Creation**

Institutions shall register their organization through the platform's institutional onboarding interface.

**4.2.2 SSO Integration**

Institutions shall be able to configure single sign-on authentication through supported identity providers.

**4.3 User Profile Creation**

**4.3.1 Profile Information Collection**

Users shall provide profile details including academic institution, specialization, and learning preferences.

**Phase 2 – Learning Personalization**

**4.4 Learning Data Tracking**

**4.4.1 Learning Progress Tracking**

The platform shall track modules accessed, modules completed, and time spent on simulations.

**4.4.2 Assessment Performance Tracking**

Student assessment scores shall be recorded and linked to the learner profile.

**4.5 Recommendation Engine**

**4.5.1 Personalized Module Recommendations**

The platform shall recommend modules based on the user's learning history and performance.

**4.5.2 Certification Path Recommendations**

Students shall receive recommendations for certification programs aligned with their learning progress.

**SECTION 5 – LEARNING EXPERIENCE FEATURES**

This section defines features that support learning interactions including assessments and AI learning assistance.

**Phase 1 – Assessment System**

**5.1 Quiz Management**

**5.1.1 Question Bank Upload**

Faculty shall be able to upload quiz question banks through structured spreadsheet templates.

**5.1.2 AI Question Generation**

Faculty shall be able to generate assessment questions using AI tools integrated into the platform.

**5.2 Assessment Types**

**5.2.1 Multiple Choice Questions**

The system shall support standard MCQ-based assessments.

**5.2.2 True/False Questions**

The system shall support binary true/false assessment questions.

**5.2.3 Descriptive Questions**

The system shall support short descriptive answer questions.

**5.3 Quiz Configuration**

**5.3.1 Randomized Question Order**

Quiz questions may be randomized for each attempt.

**5.3.2 Negative Marking**

Faculty shall be able to enable negative marking for incorrect responses.

**Phase 2 – Simulation-Based Assessment**

**5.4 XR Assessment Integration**

**5.4.1 Simulation Performance Scoring**

Scores generated during XR simulations shall be mapped to student performance records.

**5.4.2 Simulation Attempt Tracking**

The system shall track the number of simulation attempts made by each learner.

**Phase 3 – AI Learning Support**

**5.5 AI Learning Assistant**

**5.5.1 Concept Explanation Support**

The AI assistant shall provide explanations for engineering concepts related to simulation modules.

**5.5.2 Practice Question Generation**

The AI assistant shall generate practice questions for students preparing for assessments.

**SECTION 6 – LICENSING & MONETIZATION**
This section defines the licensing models used for individual learners and institutional customers.

**Phase 1 – Licensing Models**

**6.1 B2C Licensing**
**6.1.1 Individual Module Purchase**
Users shall be able to purchase individual simulation modules.
**6.1.2 Bundle Purchases**
Users shall be able to purchase bundles of multiple modules.

**6.2 Institutional Licensing**
**6.2.1 Stream-Based Licensing**
Institutions shall purchase packages mapped to academic streams.
**6.2.2 Capped Institutional License**
Institutions may purchase capped licenses based on maximum student count.

**Phase 2 – Package Management**
**6.3 Package Configuration**
**6.3.1 Bundle Creation**
Super Admin shall be able to create bundles of simulation modules.
**6.3.2 Pricing Configuration**
Pricing rules shall be configurable through the admin dashboard.

**6.4 Coupon Management**
**6.4.1 Coupon Generation**
Administrators shall be able to generate coupon codes for promotional campaigns.
**6.4.2 Coupon Expiry Configuration**
Coupons shall include configurable expiration dates.

**Phase 3 – License Lifecycle Management**

**6.5 License State Management**
**6.5.1 License Activation**
Licenses shall activate automatically after successful payment.
**6.5.2 Grace Period**
Institutional licenses shall support a configurable grace period after expiry.

## SECTION 7 – ENTERPRISE INTEGRATIONS

This section defines how the platform integrates with institutional systems such as ERP platforms and identity providers. These integrations ensure automated user provisioning, student enrollment synchronization, and seamless authentication across institutional environments.

### Phase 1 – Core ERP Synchronization

Phase 1 establishes baseline ERP synchronization to allow institutions to automatically provision student accounts and assign simulation access based on academic programs.

### 7.1 ERP Integration Framework

#### 7.1.1 Supported ERP Systems

The platform shall support integration with widely used academic ERP systems including Oracle ERP, Moodle-based LMS environments, and other ERP platforms through standardized API integration.

#### 7.1.2 ERP API Integration

The platform shall provide API endpoints that allow external ERP systems to push student enrollment data, course mappings, and academic structure information into the LMS.

### 7.2 Student Auto-Provisioning

#### 7.2.1 Scheduled ERP Synchronization

The system shall support scheduled synchronization with institutional ERP systems at configurable intervals to update student records and course assignments.

#### 7.2.2 ERP-Based Account Creation

Student accounts shall be automatically created when student records are received from ERP systems during synchronization.

#### 7.2.3 Stream-Based Access Assignment

Students shall be automatically assigned simulation packages based on the academic stream information received from the ERP system.

### Phase 2 – Real-Time Provisioning

Phase 2 introduces real-time synchronization capabilities to allow instant account creation and access assignment.

### 7.3 Real-Time ERP Provisioning

#### 7.3.1 First Login Provisioning

When a student logs into the platform for the first time, the system shall trigger a real-time API request to the ERP system to retrieve the latest academic profile.

#### 7.3.2 Dynamic Entitlement Assignment

Simulation module access shall be automatically assigned based on the course mapping received from the ERP system.

### Phase 3 – Enterprise Identity Integration

Phase 3 introduces enterprise-grade authentication capabilities.

### 7.4 Single Sign-On Integration

#### 7.4.1 SSO Authentication Support

The platform shall support institutional single sign-on authentication through providers such as Google Workspace, Microsoft Azure Active Directory, and other identity providers.

#### 7.4.2 Role-Based Access Mapping

User roles such as Student, Faculty, and Institute Admin shall be automatically mapped based on identity provider attributes.

## SECTION 8 – ADMINISTRATION & ROLE MANAGEMENT

This section defines the administrative control structure of the platform including role definitions and access permissions for different stakeholders.

### Phase 1 – Core Administrative Roles

Phase 1 establishes the foundational role hierarchy required to manage the platform.

#### 8.1 Role Hierarchy

##### 8.1.1 Super Admin Role

The Super Admin role shall have full administrative control over platform configuration, module publishing, licensing management, and cross-tenant analytics.

##### 8.1.2 Content Manager Role

The Content Manager role shall manage simulation module uploads, module metadata updates, and content readiness for publishing.

##### 8.1.3 Institute Admin Role

The Institute Admin role shall manage institutional configurations including course structures, student access, and institutional reporting.

##### 8.1.4 Faculty Role

Faculty users shall manage course assignments, assessments, and student performance monitoring.

##### 8.1.5 Student Role

Students shall access licensed modules, complete simulations, and view personal learning progress.

### Phase 2 – Administrative Automation

Phase 2 introduces administrative automation tools.

#### 8.2 Institutional Analytics Dashboard

##### 8.2.1 Institutional Usage Monitoring

Institute Admins shall have access to dashboards displaying module usage, student engagement metrics, and completion statistics.

##### 8.2.2 License Utilization Tracking

Administrators shall be able to monitor license utilization to determine whether institutional licenses are fully utilized.

### Phase 3 – Custom Role Configuration

Phase 3 introduces flexible administrative permissions.

#### 8.3 Dynamic Role Configuration

##### 8.3.1 Custom Role Creation

Administrators shall be able to define custom roles with specific permissions tailored to institutional requirements.

##### 8.3.2 Permission-Level Access Control

Custom roles shall support granular permission assignment across content management, reporting, and administrative actions.

## SECTION 9 – PAYMENT & BILLING

This section defines how payments are processed and how billing is managed for both individual and institutional customers.

### Phase 1 – Core Payment Processing

Phase 1 establishes the payment infrastructure required to support purchases and subscription management.

#### 9.1 Payment Gateway Integration

##### 9.1.1 Indian Payment Gateway Support

The platform shall support integration with major Indian payment gateways to allow domestic transactions.

##### 9.1.2 International Payment Gateway Support

The platform shall support international payment gateways to allow global purchases by individual learners.

#### 9.2 License Activation

##### 9.2.1 Automated License Activation

Upon successful payment, licenses shall be automatically activated and linked to the user account.

##### 9.2.2 Entitlement Assignment

Purchased modules or packages shall be automatically assigned to the user's dashboard.

### Phase 2 – Institutional Billing

Phase 2 introduces billing capabilities required for institutional subscriptions.

#### 9.3 Institutional Invoicing

##### 9.3.1 GST-Compliant Invoice Generation

Invoices generated for institutional purchases shall comply with GST regulations.

##### 9.3.2 Institutional Subscription Billing

The platform shall support annual institutional subscriptions with invoice generation and payment tracking.

### Phase 3 – Advanced Billing Controls

Phase 3 introduces subscription lifecycle management.

#### 9.4 Payment Monitoring

##### 9.4.1 Payment Status Tracking

Administrators shall be able to track payment status for institutional accounts.

##### 9.4.2 Grace Period Notifications

Institutions shall receive automated notifications when license renewal deadlines approach.

## SECTION 10 – SECURITY & COMPLIANCE

This section defines the security mechanisms implemented to protect intellectual property, user data, and system integrity.

### Phase 1 – Core Platform Security

Phase 1 establishes the essential security infrastructure.

#### 10.1 Asset Protection

##### 10.1.1 XR Asset Encryption

All XR simulation asset bundles shall be encrypted before distribution to prevent unauthorized access.

##### 10.1.2 Secure Module Authorization

Modules shall require license validation before execution within the XR application.

#### 10.2 User Authentication

##### 10.2.1 Secure Login Authentication

The platform shall enforce secure authentication protocols to verify user identities before granting access.

##### 10.2.2 Session Management

User sessions shall automatically expire after inactivity to prevent unauthorized access.

#### 10.3 Vulnerability Assessment

##### 10.3.1 VAPT Compliance

The platform shall undergo periodic Vulnerability Assessment and Penetration Testing to identify security risks.

##### 10.3.2 Security Patch Management

Identified vulnerabilities shall be addressed through timely security patches.

### Phase 2 – Compliance Framework

Phase 2 introduces advanced compliance capabilities.

#### 10.4 Data Protection Compliance

##### 10.4.1 Data Access Controls

Role-based access controls shall restrict access to sensitive user data.

##### 10.4.2 Data Retention Policies

The platform shall implement configurable data retention policies for user data and analytics records.

**SECTION 11 – ANALYTICS & REPORTING**

This section defines the analytics infrastructure used to track learning engagement and platform usage.

**Phase 1 – Core Learning Analytics**

**11.1 Learning Activity Tracking**

**11.1.1 Module Completion Tracking**

The system shall record module completion status for each learner.

**11.1.2 Time Spent Tracking**

The platform shall track the time spent by students interacting with simulations.

**11.2 Performance Analytics**

**11.2.1 Assessment Score Tracking**

Assessment scores shall be stored and associated with the learner's academic profile.

**11.2.2 Attempt Tracking**

The system shall record the number of attempts made for each simulation or assessment.

**Phase 2 – Institutional Reporting**

**11.3 Institutional Analytics**

**11.3.1 Class-Level Performance Reports**

Faculty shall be able to view aggregated performance metrics for their classes.

**11.3.2 Institutional Adoption Reports**

Institute Admins shall receive reports on student engagement and module adoption.

**Phase 3 – Advanced XR Analytics**

**11.4 XR Simulation Analytics**

**11.4.1 Simulation Heatmaps**

The system shall generate heatmaps identifying areas where learners frequently encounter errors.

**11.4.2 Decision Path Analysis**

Simulation decision paths shall be analyzed to identify common mistakes and learning patterns.

## SECTION 12 – PLATFORM INFRASTRUCTURE & HOSTING

This section defines the infrastructure used to host and scale the platform.

### Phase 1 – Core Cloud Infrastructure

#### 12.1 Cloud Deployment

##### 12.1.1 Cloud Infrastructure Hosting

The platform shall be deployed on cloud infrastructure providers such as AWS, Azure, or Google Cloud.

##### 12.1.2 Encrypted Asset Storage

XR asset bundles shall be stored in encrypted cloud storage systems.

#### 12.2 Content Distribution

##### 12.2.1 CDN-Based Asset Delivery

Simulation assets shall be delivered through a Content Delivery Network to reduce download latency.

### Phase 2 – Platform Scalability

#### 12.3 Auto-Scaling Infrastructure

##### 12.3.1 Dynamic Server Scaling

Application servers shall automatically scale based on platform traffic.

##### 12.3.2 Load Balancing

Traffic shall be distributed across multiple servers to maintain platform stability.

### Phase 3 – Global Deployment

#### 12.4 Multi-Region Infrastructure

##### 12.4.1 Regional CDN Nodes

Regional CDN nodes shall be deployed to support global user access.

##### 12.4.2 Multi-Region Database Replication

Database replication shall ensure high availability across geographic regions.